# The Poverty and Distributional Impacts of Carbon Pricing: Channels and Policy Implications

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## Introduction

The 2015 Paris Agreement marked a milestone in the international community's response to climate risks, with the key objective of limiting future global warming to between 1.5°C and 2°C above preindustrial levels (UNFCCC 2016, 2018). But progress on carbon pricing[1]—one of the primary tools that countries use to combat climate change—has been slow (World Bank 2022), with the average price of emissions worldwide currently being only about US$6 per ton of carbon dioxide ($CO_2$), well below what is needed to reach country emission pledges (Black, Parry, and Zhunussova 2022). Among the main barriers to ambitious carbon pricing reforms, the concerns about the impacts of carbon pricing reforms on poverty and inequality and the associated political ramifications have been well documented in the literature (Coady, Parry, and Shang 2018; Klenert et al. 2018; Carattini, Kallbekken, and Orlov 2019).

Designing effective and efficient policy to mitigate the poverty and distributional impacts of carbon pricing requires a full understanding of how carbon pricing affects households and workers as well as their communities. Here, the poverty impact refers to the decline in the level of consumption/income of vulnerable households even if the decline is uniform

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[1]Carbon pricing typically refers to market instruments, such as a carbon tax or an emissions trading system, that put a price on carbon emissions. According to the United Nations Framework Convention on Climate Change (UNFCCC), carbon pricing curbs greenhouse gas emissions by placing a fee on emitting and/or by offering an incentive for emitting less. While the focus of this paper is on carbon pricing, many of the channels discussed in the paper may also apply to the poverty and inequality impacts from a regulatory approach.

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across all households, and the distributional impact refers to the decline relative to that of higher-income households. Policy analyses often focus on only the impact of carbon pricing directly on the prices of energy products and indirectly on the prices of other consumption goods and services that use energy products as inputs, with simplifying assumptions on pass-throughs of energy prices and on demand responses by consumers. But a growing literature assesses how the effects of carbon pricing on employment and income as well as on health outcomes differ by population groups; further attention has also been paid to the consumption channel (e.g., how demand responses by consumers may differ by household income and other household characteristics; see Rausch, Metcalf, and Reilly 2011; Beck et al. 2015; Moshiri and Santillan 2018; Goulder et al. 2019; Hille and Möbius 2019; Hsiang, Oliva, and Walker 2019; Marin and Vona 2019).

This article provides a comprehensive review of the channels through which carbon pricing may affect poverty and inequality, surveying both theoretical and empirical literature. The framework includes the following four main impact channels: consumption, income, health, and revenue recycling. Another contribution of this article is that it differentiates channels along several dimensions (timing, place, nature, heterogeneity, and uncertainty) and discusses their implications for policy designs. Particularly, this article complements Fullerton (2011), who discusses six broad types of distributional effects by using a simple demand-and-supply framework. Several recent review papers focus on specific aspects of the distributional impacts of carbon pricing. For example, Fullerton and Muehlegger (2019) illustrate how theory of tax incidence can be applied to environmental taxes and nontax environmental regulations while taking into account various market conditions, Pizer and Sexton (2019) focus mostly on the static impacts of energy taxes and emphasize the importance of the variation in distributional impacts within income groups, and Hsiang, Oliva, and Walker (2019) assess the distribution of environmental damages.

This review indicates that despite the rapid development in the literature, significant gaps remain, particularly related to leakages through trade and exemptions, firm responses to carbon pricing, and structural changes in factor income. Furthermore, the channels differ in the timing, place, nature, heterogeneity, and uncertainty of their impact. As such, a blanket assessment of whether carbon pricing is progressive or regressive lumps together the effects from channels with different attributes and is not informative in designing effective and efficient policy responses.

The review suggests that policy designs should focus on the channels that are most relevant for the given country or region, that have near-term impacts, and that are better understood and whose impacts less uncertain. In addition, policy designs should go beyond the averages and provide support to the specific groups that are particularly affected. As climate policy design cannot wait for us to close knowledge gaps and countries must act now, a reasonable starting point is to base the analysis of the poverty and distributional impacts on the simplifying assumptions of full pass-through, no behavioral responses by firms and households, a closed economy with carbon pricing applying to all sectors, and no income effect. To this end, evidence suggests that in most cases, targeted/universal cash transfers for households and targeted assistance to particularly affected workers could fully protect low-income and vulnerable populations from carbon pricing reforms, while leaving a significant share of carbon revenues to improve economic efficiency.

The rest of the article is organized as follows: The next section starts with some considerations in defining the baseline. Then, the article systematically reviews the potential channels through which carbon pricing can affect poverty and inequality, analyzes their differences along several dimensions, and identifies key gaps in the literature. And last, the article draws lessons from country experiences and concludes with some considerations on how policy designs should account for the attributes of the channels in mitigating the impacts of carbon pricing reforms on households.

## Defining the Baseline

A clearly defined baseline helps identify the drivers of the poverty and distributional impacts under reform scenarios and compare results across studies. The baseline is typically defined in two ways in the literature. Some studies use what actually happened in the recent past—typically the year when the most recent household survey was completed—and estimate the poverty and distributional impacts had a carbon pricing scheme been introduced (Rausch, Metcalf, and Reilly 2011; Dorband et al. 2019; Wang et al. 2019). Another line of studies builds an explicit baseline, using the latest household survey and projecting into the future (Goulder et al. 2019; IMF 2019). The latter is more realistic and policy relevant as a forward-looking exercise, but it requires assumptions on the trends of key factors related to consumption and production.

In addition, the interactions between certain baseline characteristics and carbon pricing mean that the baseline itself also plays a role in the poverty and distributional impacts of carbon pricing, particularly related to carbon intensity of consumption goods and services and to access to energy. Studies find that carbon pricing tends to be progressive in developing economies, largely because lower-income households have limited access to energy, particularly electricity (Dorband et al. 2019; IMF 2019; Pizer and Sexton 2019). As countries develop, these households would gain access to energy, which should be explicitly built into the baseline and would likely worsen the poverty and distributional impacts of carbon pricing.

The development and deployment of low-carbon technologies and fuel switching would lead to smaller increases in the prices of consumption goods and services from carbon pricing and to smaller poverty and distributional impacts. Carbon intensity of consumption goods and services has declined substantially over the past decades, though the literature finds that only a small share may be attributed to the introduction of carbon pricing schemes, such as the European Union Emissions Trading System (EU ETS). Having explicit assumptions on the carbon intensity of goods and services in the baseline is thus essential to fully take into account the declining trend in carbon intensity.

## The Poverty and Distributional Impacts of Carbon Pricing: Four Channels

Carbon pricing reforms can have broad effects on the economy, including on consumption, investment, structure of the economy, health outcomes, and ultimately the welfare of households. Many policy analyses focus on consumption and make the simplifying assumptions that carbon pricing is fully passed through to consumers, that there are no behavioral responses by firms and households, and that the economy is closed, with carbon pricing applying to all sectors. The methodology is easy to understand and implement. After the carbon intensity of each energy product is determined, an input–output table can be used to trace the carbon intensity of other goods and services. Then, it is straightforward to calculate the final prices of all consumption goods and services with carbon pricing, assuming full pass-through. The overall poverty and distributional impacts depend on the budget shares of consumption goods and services for any given household and on how carbon revenues are recycled.

However, a growing literature suggests that the simplistic approach may come to incorrect conclusions on the poverty and distributional impacts of carbon pricing by ignoring important channels. This section identifies four main impact channels—consumption, income, health, and revenue recycling—and reviews available evidence on each from theoretical and empirical literature.[2]

## The Consumption Channel

### Pass-throughs

Not all the burden of carbon pricing may be passed forward in higher prices for households; some may be passed backward in lower prices for firms.[3] As a result, owners of capital and workers will bear some of the burden. The pass-through may sometimes be larger than 1 as a result of imperfectly competitive product markets with very convex demand (Weyl and Fabinger 2013; Ganapati, Shapiro, and Walker 2020). In such a case, firms' gains from over pass-through would have poverty and distributional implications through the income channel.

This is an area where empirical evidence is still emerging, and findings thus far have been mixed. Kotchen (2021) finds an average pass-through of 0.85 for coal, natural gas, gasoline, and diesel, based on demand-and-supply elasticities. Many studies use firm- or industry-level data to estimate pass-throughs, and the estimates range from below 1 to above 1, depending on sector and local market conditions (Marion and Muehlegger 2011; Fabra and Reguant 2014; Stolper 2016; Miller, Osborne, and Sheu 2017; Ganapati, Shapiro, and Walker 2020). For example, Ganapati, Shapiro, and Walker (2020) find that, on average, around 70 percent of energy price–driven changes in input costs get passed through to consumers over the short to medium term, with large heterogeneities across sectors, ranging from 0.36 (gasoline) to 1.87 (cement). Fabra and Reguant (2014) find that costs imposed by the EU ETS are almost fully passed through to consumers in the form of higher electricity prices. Interestingly, Stolper (2016) finds that pass-through rises monotonically with area-average house prices, while Harju et al. (2022) find that the effect of fuel carbon taxes on consumer prices decreases with areal income and with the degree of urbanization, which could have important implications on the poverty and distributional impacts of carbon pricing.

[2]A simple utility maximization framework can help facilitate the discussions of the channels in a systematic manner (see the appendix, available online, for details).
[3]See the appendix for an illustration.

### Production responses by firms

Conceptually, carbon pricing would incentivize firms to improve energy efficiency and to switch to less carbon-intensive energy (Jaffe, Newell, and Stavins 2002; Acemoglu et al. 2012). As a result, this can help mitigate the effects of carbon pricing on production costs and consumer prices.[4] In addition, firms' production responses could alter factor prices for labor and capital and have poverty and distributional implications through the income channel as well.

Improving energy efficiency is essentially a substitution of nonenergy input for energy input to lower production costs. This can be achieved through both more widespread adoption of existing technologies and development of new technologies. The literature indicates that firms have been employing new machinery and other equipment and making behavioral and process changes, some of which might simply not have been economically viable in the absence of carbon pricing schemes (Petsonk and Cozijnsen 2007; Tomás et al. 2010; Anderson, Convery, and Di Maria 2011). The literature also has identified nonprice barriers (including inadequate information, high uncertainty, principal–agent problems, and constrained capital financing) as important factors in determining the diffusion of low-carbon technologies. Thus, policies to address the associated market failures may complement carbon pricing reforms and help accelerate the diffusion process (Jaffe, Newell, and Stavins 2002).

Studies also find evidence of low-carbon technological innovation from existing ETS schemes (e.g., EU ETS and China ETS pilots), but their overall impact appears to be limited (Calel and Dechezleprêtre 2016; Zhu et al. 2019; Lilliestam, Patt, and Bersalli 2021). However, Aghion et al. (2016) find that the effect of carbon taxes on low-carbon innovation is significant when inferred from energy price changes or pollution taxes. The different findings may reflect low coverage, excessive allowance allocation, and low carbon prices of the ETS schemes (Calel and Dechezleprêtre 2016; Lilliestam, Patt, and Bersalli 2021).

In addition to improving energy efficiency, carbon pricing can provide incentives for firms to switch toward less carbon-intensive energy, such as from coal to natural gas or renewables in electricity production. The literature has documented fuel switching, particularly by energy firms, and shows that it accounts for the majority of the carbon emissions reduction under the EU ETS (Anderson, Convery, and Di Maria 2011; Borghesi et al. 2015; Calel and Dechezleprêtre 2016; Lilliestam, Patt, and Bersalli 2021).

Firms' production responses could have large impacts on production costs and consumer prices in the short and long terms, which would deliver a bigger benefit to consumers whose consumption is more carbon intensive under the baseline. Sager (2019) suggests that the substitution of intermediate inputs along global value chains and fuel switching can significantly mitigate the consumer price increases from carbon pricing, even without taking into account the impact on energy-saving technological innovation and the substitution between energy and other inputs.

### Leakages

The discussions thus far assume a closed economy. International trade can further reduce the impact of carbon pricing schemes on consumer prices. In the absence of border carbon adjustments or some form of global carbon pricing, imported intermediate inputs or final

[4] See the appendix for an illustration.

consumption goods and services would not be subject to carbon pricing, and their prices could remain at the levels prior to the introduction of carbon pricing. Furthermore, some foreign goods and services whose prices were previously not competitive may further replace domestic production. Separately, ETS schemes often cover only large emitters, and even carbon tax schemes sometimes provide exemptions for selected sectors, which would also lead to lower consumer prices.

### Demand responses by consumers

In addition to firms' behavioral responses, consumers' behavioral responses to carbon pricing can further mitigate its impact on household welfare. Consumption level and composition will respond to changes in price levels and relative prices as a result of carbon pricing. West and Williams (2004) find that ignoring demand responses could substantially overstate the impact on consumers and that the differences between equivalent variation and easier-to-implement consumer surplus measures are relatively small. Sager (2019) suggests that the emissions reduction from demand responses would likely be moderate. A carbon price of US$30 per ton of $CO_2$ would reduce carbon emissions by about 14 percent, mostly because of consumers substituting away from emissions-intensive goods rather than lowering consumption in response to across-the-board price increases. In addition, behavioral responses may vary across income groups (Moshiri and Santillan 2018; Muller and Yan 2018). However, empirical estimates appear to vary substantially across studies because of differences in methodologies and data (Zhu et al. 2019).

The rebound effect—the phenomenon that an improved energy efficiency can lead to an increase in energy use because the cost of the energy service declines—has been extensively discussed in the context of energy efficiency, and it needs to be taken into account when estimating consumer responses in both the baseline and the carbon pricing scenario. The effect appears significant with large variations across studies, depending on sector and type of efficiency improvement (Frondel, Ritter, and Vance 2012; Gillingham et al. 2013).

## The Income Channel

### Destruction of brown jobs

Substantive carbon prices could have large negative impacts on certain groups of workers and regions engaged in carbon-intensive activities. For example, a carbon tax could lead to substantial job losses in the coal sector (Morris 2016; IMF 2019). Typically, coal-related (or fossil fuel–related) jobs are highly geographically concentrated, accounting for a large share of local employment. Winding down production in these regions could lastingly reduce output and employment prospects for local communities. In addition, extractive activities may cause scarred local landscapes and impaired waterways, reducing prospects for attracting new industries (Morris 2016). Although carbon-intensive sectors often account for only a relatively small share of overall employment, the impacts of carbon pricing on workers in these sectors and the associated policy responses are of great relevance in the political economy of carbon pricing reforms in many countries. A better understanding of the employment and wage impacts of carbon pricing for workers in these sectors is particularly important. Walker (2013) studies environmental regulations and finds that the average worker in a regulated sector experienced a total earnings loss equivalent to 20 percent of their preregulatory earnings and that almost all the estimated earnings losses are driven by workers who separate from their firm.

A gradual increase of carbon tax to US$50 per ton of $CO_2$ by 2030 could substantially accelerate already-declining coal production and coal-related employment relative to 2015 levels. For example, in the United States, coal-related job losses would jump from 8 to 55 percent. In China, coal-related jobs would decline by 45 percent instead of by a small loss; in India, where coal-related employment is expected to grow by 5 percent, it would decline by 42 percent with the carbon tax. These job losses would amount to 0.3–0.9 percent of economy-wide employment in China and Poland and to less than 0.15 percent in other countries (figure 1).

On a larger scale, global action on climate change could have a substantial economy-wide impact on countries that heavily rely on oil income for their economic growth and fiscal revenues (IMF 2019). This reduction in revenues would need to be compensated through either tax increases or expenditure reductions. The exact poverty and distributional impacts would thus depend on the designs of the fiscal adjustment measures.

## Creation of green jobs

Carbon pricing may lead to an increase in employment in other sectors as a result of, for example, investments in less carbon-intensive energy, such as renewables. With carbon pricing,

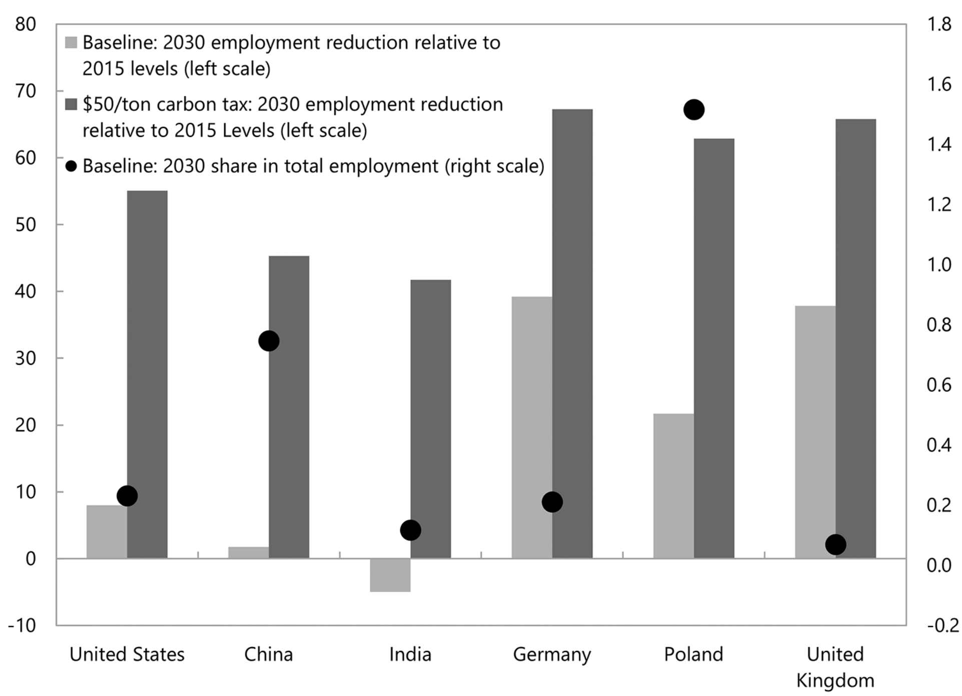


**Figure 1** Impact of carbon pricing on employment in the coal sector (carbon tax of US$50/ton of $CO_2$; IMF 2019). A color version of this figure is available online.

renewables would become more economically viable, and investments and employment would likely increase in the sector (Yamazaki 2017; Hille and Möbius 2019; IMF 2019). Yamazaki (2017) finds that British Columbia's carbon tax has a modest positive impact on employment. While the aggregate impact is small, the paper finds significant job shifting from carbon-intensive to non-carbon-intensive sectors.

### Income gain due to climate improvements

Lower-income households potentially could reap greater benefits from carbon pricing schemes than higher-income households to the extent that carbon pricing can help reduce the frequency and severity of damaging climate events, such as floods, droughts, and tropical cyclones—for several reasons (World Bank 2004; Hsiang, Oliva, and Walker 2019). First, marginal damages from climate changes tend to be larger for lower-income populations because, for example, they may have less access to credit or technology and thus are less capable to insure themselves or build resiliency against adverse climate events. Second, lower-income populations may derive a greater proportion of their income from sectors that are most vulnerable to climate change, such as agriculture, forestry, and fisheries. Third, lower-income populations may be located in areas that are more prone to climate-related damage, but empirical evidence appears mixed on this.

### Structural changes in factor income and demand for skills

Studies suggest that low-carbon structural changes from carbon pricing may have direct distributional implications. First, carbon pricing may alter the demand for workers with different skill levels. Marin and Vona (2019) find that the skill bias of climate policies mostly consists of a substitution of technical and, to a lesser extent, professional workers for manual workers. Second, to the extent that the pass-through from carbon pricing is not unity, the burdens or benefits from carbon pricing may differ by capital and labor (Rausch, Metcalf, and Reilly 2011). Third, if carbon-intensive industries are also capital intensive, and if carbon can be more easily substituted by labor than by capital, returns to capital would decline more than wages; as such, households with greater income shares derived from capital would suffer more. This suggests that the composition of income (from labor, capital, and government transfers) by income level is an important determinant of the distribution impact of carbon pricing reforms (Fullerton and Heutel 2010; Rausch et al. 2010; Dissou and Siddiqui 2014; Beck et al. 2015; Goulder et al. 2019).

A number of studies use general equilibrium models to explore the distributional implication of the factor income channel. One main takeaway from this literature is that the factor income channel could be important, and in some cases, it appears sufficient to reverse the overall distributional impact of carbon pricing reforms from regressive to progressive. The assumptions that government transfers—which are not part of carbon revenue recycling—are assumed to remain unchanged in real terms from the baseline to the carbon pricing scenario and are counted as part of household income may contribute to this result, however (Rausch et al. 2010; Rausch, Metcalf, and Reilly 2011; Dissou and Siddiqui 2014; Beck et al. 2015; Cronin, Fullerton, and Sexton 2019; Goulder et al. 2019). Another takeaway is that the results tend to be sensitive to parameter value assumptions, for example, on factor intensities and factor substitution rates, highlighting the need for further empirical evidence to support the parameterization of these models (Fullerton and Heutel 2010; Cao, Ho, and Ma 2020).

## The Health Channel

According to the World Health Organization, 4.2 million deaths every year occur as a result of exposure to ambient (outdoor) air pollution. Carbon pricing can lead to substantial improvements in air quality and help reduce such deaths (Coady et al. 2017; Karlsson, Alfredsson, and Westling 2020; Parry et al. 2020).

Some indicative evidence suggests that the health cobenefits may vary by population groups. For example, empirical studies have documented that low-income and vulnerable households disproportionately live in areas with higher exposure to air pollution (Boyce and Pastor 2013; Zwickl, Ash, and Boyce 2014; EEA 2018; Hsiang, Oliva, and Walker 2019). Tessum et al. (2019) show that in the United States, $PM_{2.5}$ exposure is disproportionately caused by consumption of goods and services mainly by the non-Hispanic white majority but is disproportionately inhaled by black and Hispanic minorities. The empirical evidence on how carbon pricing affects environmental disparities is still unfolding. Cushing et al. (2018) find that in the first three years of California's cap-and-trade program, environmental disparities in industrial air pollution exposure were not reduced, while Hernandez-Cortes and Meng (2021) find that the program has narrowed environmental disparities since 2013.

Additionally, carbon pricing is expected and has been found to reduce traffic-related injuries and fatalities (Grabowski and Morrisey 2004; Parry, Walls, and Harrington 2007; Burke and Nishitateno 2015). The distributional impact is not well studied, however, and it involves a complex mix of actors related to both the physical harm and the associated medical treatment and compensation, including through insurance.

## The Revenue Recycling Channel

Revenue recycling is an essential part of a carbon pricing reform package, which typically consists of elements to improve efficiency and support growth (e.g., tax cuts and public investment) and elements to reduce poverty and inequality (e.g., transfers to households). Intended or not, all these elements could have significant poverty and distributional implications (Klenert et al. 2018; IMF 2019; Metcalf 2019).

The poverty and distributional impacts of revenue recycling measures often depend on their exact designs. For personal income tax, it depends on whether the tax cut is at the top income bracket or at the bottom income bracket; for public spending on education and healthcare, it depends on whether the spending is targeted at low-income households, less developed regions, or poorer communities. Targeted cash transfers through social safety nets can protect the most vulnerable with lower fiscal cost—but they may also leave coverage gaps and provide disincentives to work, particularly in developing economies with limited administrative capacity.

The distributional impacts of corporate income tax cuts and public investment in infrastructure, including green investment, are less well understood. The evidence on how a corporate income tax cut would be shared between labor and capital is still mixed (Fuest, Peichl, and Siegloch 2018; Nallareddy, Rouen, and Suárez Serrato 2018). Public infrastructure investment could be progressive if targeted at less developed regions or sectors with low wages.

Empirical evidence is still scarce, but early research indicates that public investment tends to lower inequality in developing economies (Furceri and Li 2017).

While it is important to understand the poverty and distributional impacts of each tax or expenditure measure, what matters in the end is the performance of the reform package as a whole in achieving poverty and distributional objectives with the least efficiency cost. A combination of public investment and targeted cash transfers may be preferred to universal cash transfers, for example, in a country with large public investment needs. Eventually the measures to recycle carbon revenues should be integrated into the broad tax-benefit systems to achieve optimal allocation of all public resources as a whole.

## Attributes of the Impact Channels

Carbon pricing can affect household poverty and inequality through many channels, some of which are better understood than others. Table 1 summarizes the main findings from the literature. Often when studies incorporate multiple channels in their analysis, there is a tendency to lump their effects together and make a blanket statement on whether carbon pricing is progressive or regressive. However, a simple conclusion on the overall progressivity of carbon pricing masks the differences in each channel's welfare effect—potentially leading to policy designs that may inadequately protect the most vulnerable and jeopardize political support for carbon pricing reforms. Specifically, the channels can be assessed by the timing, place, nature, heterogeneity, and uncertainty of impact.

"Timing of impact" refers to how long it would take for the distributional impacts of carbon pricing to materialize once it is introduced. The impacts from some channels are almost immediate, including the pass-through of carbon pricing from firms to consumers, leakages through trade and exemptions, and revenue recycling measures, such as tax cuts and cash transfers. Other channels, such as the income gain from climate improvements, may only have an economically meaningful impact over the long term. Still others may have both short- and long-term impacts, such as production responses from firms, destruction of brown jobs and creation of green jobs, structural changes in factor income and demand for skill, and health cobenefits. This suggests that even if the combined distributional effect is progressive at steady state, it does not necessarily mean that the most vulnerable would not suffer more in the short run. Policy designs will need to take this into account.

"Place of impact" refers to the extent to which the distributional impacts of the channels concentrate on certain geographical locations. For example, job losses from the coal industry already have and will continue to have severe impacts on the regions where coal mines are located. Major oil-exporting countries will be hit hard if oil demand and oil prices fall as a result of global climate actions. These regions or countries may not be able to attract green jobs immediately, because, for example, they lack a skilled workforce. Potential constraints on labor mobility in the short run could prevent workers in the coal industry from taking advantage of job opportunities elsewhere. Therefore, even if carbon pricing does not result in net job loss, targeted policies to assist particularly affected regions may still be needed to facilitate the transition of their local economies.

"Nature of impact" refers to whether the impacts are monetary or in-kind. Most of the channels affect either household income or household expenditure, so their aggregation is

**Table 1** Overview of the channels and findings in the literature

| Main channel, subchannel | Useful references | Key findings and remaining issues |
|---|---|---|
| Consumption channel | | |
| Pass-throughs | Ganapati, Shapiro, and Walker 2020; Kotchen 2021 | The literature finds large variation in the estimates of pass-throughs across sectors. The overall impact on consumer prices is still unclear. |
| Production responses by firms | Jaffe, Newell, and Stavins 2002; Aghion et al. 2016; Calel and Dechezleprêtre 2016; Sager 2019; Lilliestam, Patt, and Bersalli 2021 | Carbon pricing could lead to a large reduction in production costs and consumer prices, as firms adopt existing low-carbon technologies, develop new low-carbon innovations, and switch fuels toward those with low-carbon content. Quantitative estimate of the overall impact, however, is still lacking. |
| Leakages, including through trade and incomplete coverage of carbon pricing schemes | Sager 2019 | Research quantifying this effect is still limited. |
| Demand responses by consumers | West and Williams 2004; Dimitropoulos et al. 2018; Muller and Yan 2018; Zhu et al. 2019 | Demand responses can help mitigate the impact of carbon pricing on households; however, to what extent such responses differ by income and other household characteristics is still unclear. |
| Income channel | | |
| Destruction of brown jobs | Walker 2013; Morris 2016; IMF 2019 | The impact can be large for certain sectors, communities, and even countries. |
| Creation of green jobs | Yamazaki 2017; Hille and Möbius 2019 | There is some evidence that carbon pricing can lead to a shift of employment from carbon-intensive to noncarbon-intensive sectors. The distributional implication of such a shift, however, is unclear. |
| Income gain due to climate improvements | Hsiang, Oliva, and Walker 2019 | Low-income households likely can benefit more from climate improvements; however, further evidence is still needed. |
| Structural changes in factor income and demand for skills | Fullerton and Heutel 2010; Rausch, Metcalf, and Reilly 2011; Beck et al. 2015; Goulder et al. 2019; Marin and Vona 2019 | Results from general equilibrium models suggest that the effects from changes in factor income could be large. Part of the results are driven by the assumption on government transfers, which are part of household income. In addition, the results may be sensitive to parameter assumptions. Research on the impact of carbon pricing on demand for skills is still scarce. |
| Health channel | | |
| Reduction in air pollution | Burke and Nishitateno 2015; Hsiang, Oliva, and Walker 2019; Parry et al. 2020; Hernandez-Cortes and Meng 2021 | The health cobenefits from reduction in air pollution is found to be substantial, and there is indicative evidence that low-income and disadvantaged households may benefit more. However, little empirical evidence is available from existing carbon pricing schemes. |
| Reduction in traffic-related injuries and fatalities | Burke and Nishitateno 2015 | There is little research on how the effect differs by population groups. |

**Table 1** (*Continued*)

| Main channel, subchannel | Useful references | Key findings and remaining issues |
|---|---|---|
| Revenue recycling channel | | |
| Tax cuts | Fuest, Peichl, and Siegloch, 2018; Nallareddy, Rouen, and Suárez Serrato 2018 | The distributional impact of a personal income tax cut depends on the design; the distributional impact of a corporate income tax cut is still being debated. |
| Boosting public investment in human capital and infrastructure | Furceri and Li 2017; Coady and Dizioli 2018 | The distributional impacts would highly depend on the design of the policies. Programs to expand access to education and healthcare are likely to be progressive. There is still limited evidence on the distributional impact of infrastructure investment, including green investment. |
| Targeted or universal cash transfers | Coady and Le 2020 | Both targeted and universal transfers can help mitigate the poverty and distributional impacts of carbon pricing but with tradeoffs in terms of fiscal cost, coverage, and work incentives. The appropriate measure would be country specific, depending also on administrative capacity. |

straightforward. However, emissions reductions can lead to important health cobenefits, and revenues raised from carbon tax or other pricing mechanisms can be invested in public health or education. Then, the challenge is how to translate the in-kind benefits into household welfare, given that their valuation may differ by population groups (Hsiang, Oliva, and Walker 2019).

"Heterogeneity of impact" refers to the variations in the impacts of carbon pricing within groups, such as those defined by prereform income. Exclusively focusing on averages across income groups may obscure important within-group variations. Rausch, Metcalf, and Reilly (2011) find large variations by race and region within income groups on both the use side—particularly the consumption of electricity, coal, and natural gas—and the source side. Just like many other reforms, carbon pricing creates winners and losers. As it reallocates jobs across sectors, the workers who lose jobs in negatively affected industries may not be the same workers who are hired in industries that increase hiring, especially in the short run. This calls for targeted policies to assist those who are negatively affected by the reform beyond what is suggested by group averages.

"Uncertainty of impact" refers to how precisely we can assess the impacts of the channels based on existing evidence. Some channels are inherently uncertain, such as the development of low-carbon technological innovations and the climate impact of carbon pricing (e.g., the frequency of extreme weather events). Knowledge gaps and data limitations can also lead to uncertainty and would require additional investment in research and development.

Table 2 highlights how the channels differ along these five dimensions. Together with table 1, it also helps identify gaps in the literature and topics for future research. A few areas that are worth emphasizing include firm responses, structural changes in factor income, and leakages.

**Table 2** Key attributes of the channels

| Main channel, subchannel | Timing | Place | Nature | Heterogeneity | Uncertainty |
|---|---|---|---|---|---|
| Consumption channel | | | | | |
| Pass-throughs | Short to medium term | All | Monetary | Small | High |
| Production responses by firms | Short to long term | All | Monetary | Small | High |
| Leakages, including through trade and incomplete coverage of carbon pricing schemes | Short term | All | Monetary | Small | Low |
| Demand responses by consumers | Short to medium term | All | Monetary | Medium | High (with respect to how demand responses vary by different groups) |
| Income channel | | | | | |
| Destruction of brown jobs | Short to long term | Areas where fossil-fuel industries are concentrated | Monetary | Large | Medium |
| Creation of green jobs | Short to long term | Not necessarily in the same areas where brown jobs are lost | Monetary | Large | High |
| Income gain due to climate improvements | Long term | Climate-vulnerable areas | Monetary | Large | High |
| Structural changes in factor income and demand for skills | Short to long term | All | Monetary | Medium | High |
| Health channel | | | | | |
| Reduction in air pollution | Short to long term | Areas that are currently most affected (e.g., urban areas and adjacent areas to coal power plants) | In-kind | Large | Medium |
| Reduction in traffic-related injuries and fatalities | Short to long term | All | In-kind | Large | Medium |
| Revenue recycling channel | | | | | |
| Tax cuts | Short term | All | Monetary | Small | Medium (with respect to corporate income tax) |
| Boosting public investment in human capital and infrastructure | Medium to long term | Targeted areas | Monetary/ in-kind | Large | High |
| Targeted or universal cash transfers | Short term | All | Monetary | Small | Low |

Note: The assessments here are illustrative given that the evidence is still emerging and that the findings often vary significantly across studies. Here, timing of impact refers to how long it would take for the distributional impacts of carbon pricing to materialize once it is introduced; place of impact refers to the extent to which the distributional impacts of the channels concentrate on certain geographical locations; nature of impact refers to whether the impacts are monetary or in-kind, with the latter raising the issues of different valuation by different groups; heterogeneity of impact refers to the variations in the impacts of carbon pricing within groups, such as those defined by prereform income or other socioeconomic characteristics; and uncertainty of impact refers to how precisely the impacts of the channels can be assessed based on existing evidence, with high uncertainty due to either intrinsic uncertainty of the channel or large variation in empirical evidence.

Existing evidence suggests that firm responses—including pass-throughs and production responses—could significantly reduce the impact of carbon pricing on consumers prices and thus help limit the welfare loss by households and the size of mitigating measures. However, research is limited, particularly in quantifying their impacts and understanding their transition paths.

In addition, a number of studies based on general equilibrium models suggest that structural changes in factor income are likely an important channel. However, the findings are very much based on theoretical models with limited empirical support, and the results are also sensitive to parameter value assumptions. Therefore, further research in the area is warranted, particularly on the empirical side to support the model parametrization and provide evidence from existing carbon pricing schemes. In addition, it may take time for its effect to materialize, and research on the transition path would be very valuable.

Another area that has been overlooked in the literature is the distributional implication of leakages. Given the challenge of coordinated global climate actions, the difficulties in implementing border carbon adjustments, and the large exemptions in many existing designs of carbon pricing schemes, this channel could significantly reduce the burden on households at the costs of carbon emissions reductions and cobenefits.

## Lessons from Country Experiences

Many countries have introduced carbon pricing, some of which have taken effect for decades. Their experiences could provide useful lessons along with the empirical literature. Table 3 summarizes the reform experiences of eight carbon pricing schemes (British Columbia, California, China, Colombia, France, Singapore, South Africa, and Sweden).

All the schemes adopted a gradual approach to carbon pricing reforms, which allows firms and households time to adjust. Carbon pricing of several recent schemes starts at very low levels, possibly reflecting uncertainty about its impacts (China, Colombia, Singapore, and South Africa).[5] In South Africa, the effective rate is only US$0.30–US$1.20 per ton of $CO_2$ because of exemptions and offset allowances (World Bank 2020).

Nearly all schemes include some measures to soften the impact on firms. This includes tax cuts (British Columbia), exemptions (Colombia, France, and South Africa), free emissions permits (California), and low initial carbon tax rates (Singapore and Sweden). However, no scheme has implemented border carbon adjustments thus far. In the case of China, while no information is available on the support given to firms, highly regulated electricity tariffs could mitigate the impact. This support would have implications on poverty and inequality through the leakage channel.

Only some countries introduced measures to support households, probably in part because of the very low effective levels of carbon pricing in some cases (China, Colombia, and South Africa). In South Africa, the carbon tax scheme will not have any impact on electricity prices in the first phase; in China, electricity tariffs are regulated.

[5]A review of South Africa's carbon tax scheme will be conducted before the second phase, which begins in 2026. Future changes to rates and tax-free thresholds in the carbon tax will follow after the review. Meanwhile, Singapore recently announced plans to raise its carbon tax from US$5 per ton of $CO_2$ to US$25 per ton in 2024–2025 and to US$45 per ton in 2026–2027, with a view of reaching US$50–US$80 per ton by 2030.

**Table 3** Country experiences in designing carbon pricing schemes

| Country | Reform description | Household support | Firm support | Revenue use |
|---|---|---|---|---|
| British Columbia | The carbon tax applies to the purchase and use of fossil fuels and covers around 70 percent of provincial greenhouse gas emissions. It was introduced on July 1, 2008, at US$10 per ton of $CO_2$ and raised to US$40 per ton on April 1, 2022. | Tax credit to offset the impact of the carbon taxes paid by low- and middle-income individuals and families, progressive personal income cuts, and tax credits and transfers targeted at various groups. | Corporate income tax cuts, particularly for small businesses; tax cuts and credits for targeted uses (e.g., training and scientific research). | British Columbia's carbon tax is designed to be revenue neutral, in the sense that all revenues raised are to be recycled to households and businesses, largely in the form of tax cuts. In practice, there have been some deviations. |
| United States (California) | California's carbon cap-and-trade program was launched in 2013. The program covers approximately 80 percent of California's greenhouse gas emissions, including large electric power plants, large industrial plants, and fuel distributors. | At least 25 percent of revenues go to projects within and benefitting disadvantaged communities, and at least an additional 10 percent is for low-income households or communities. Examples include energy bill rebates and home weatherization programs. | Regulated firms receive most of their emissions permits free of charge, the compliance offset program provides important cost containment and compliance flexibility for covered entities, and a cost containment mechanism known as the Allowance Price Containment Reserve helps prevent allowance prices from rising too high. | Revenues from the program are deposited into the state's Greenhouse Gas Reduction Fund and used to implement programs that further reduce greenhouse gas emissions. At least 35 percent of the revenues are required by law to be directed to environmentally disadvantaged and low-income communities. |
| China | China's national ETS, launched in 2021, initially covers only the power sector. Its coverage is expected to be gradually expanded to other sectors. | No information is available. However, electricity tariffs in China are highly regulated. | No information is available on any support for power-sector firms. For other firms, electricity tariffs in China are highly regulated. | No revenue gain due to free allowance allocation. |
| Colombia | An economy-wide carbon tax at US$5 per ton on all liquid and gaseous fossil fuels used for combustion was introduced in 2017 and planned to be gradually increased to US$11 per ton. | No information is available. | Exemptions apply to natural gas consumers that are not in the petrochemical and refinery sectors and to fossil fuel consumers that are certified to be carbon neutral. | Revenue from Colombia's carbon tax is earmarked for the Colombia in Peace Fund, which supports activities such as watershed conservation, ecosystem protection, and coastal erosion management. |
| France | Carbon tax was introduced on non-ETS emissions in 2014. Rates were initially set at US$8 per ton, rose to | A compensation scheme was introduced in 2015 to provide financial assistance to low- | Agriculture, taxis, and trucks are exempted from the carbon tax to protect their competitiveness. | While, in general, France does not earmark revenues, the reform was accompanied by some |

| | | | | |
|---|---|---|---|---|
| | US$36 per ton in 2017, and were on a trajectory to reach US$97 per ton in 2022. Ramping up of tax was suspended at around US$50 per ton in 2018. | income households on their energy bill. | | support to energy transition, financial assistance to low-income households, and broad tax reductions. |
| Singapore | Carbon tax, applying to all large emitters and covering about 80 percent of total emissions, is set at US$5 per ton from 2019 to 2023, with plans to increase it to around US$25 per ton in 2024–2025, US$45 per ton in 2026–2027, and US$50–US$80 per ton by 2030. | Improving energy efficiency of public housing, supporting low-income households in purchasing more energy efficient appliances, and ensuring that consumers are not overcharged by electricity retailers. | Starting at a low level takes into account the potential impact on competitiveness. | Support initiatives to address climate change, such as incentives for energy efficiency improvements in the industrial sector. |
| South Africa | There are two phases: 2019–2025 and after 2025. The carbon tax rate was US$7 per ton in 2020, and it is planned to reach US$20 per ton by 2026 and then to increase more rapidly thereafter to at least US$30 by 2030 and up to US$120 beyond 2050. The tax applies to industry, power, buildings, and transport sectors, irrespective of the fossil fuel used. | The introduction of the carbon tax will not have any impact on the price of electricity for the first phase. | Exemptions and offset allowances vary by sector (e.g., trade-exposed companies receive additional allowances). Companies could receive tax-free allowances ranging from 60 to 95 percent of their emissions, reducing the effective tax rate to US$0.30–US$1.20 per ton. | The tax will go to the general budget. There is indication that most carbon tax revenues will be used to prevent an increase in the price of electricity during the first phase of the carbon tax. Any leftover revenues will fund new energy efficiency initiatives via the budget process. |
| Sweden | Carbon tax on motor fuels and heating fuels was introduced in 1991 at US$28 per ton (industries covered by the EU ETS are excluded) and increased to US$130 per ton by 2022. Lower rate for industry (at US$7 per ton in 1991) was phased out by 2018. | Social transfers and increased basic income tax reductions for low- and middle-income households were adopted to address undesirable distributional consequences of the carbon tax. | A lower initial rate for industry (at US$7 per ton) and the adjustments during following years take into account competitiveness concerns. | Carbon tax revenues go to the general budget. General budget funds may be used for specific purposes linked to the carbon tax, such as addressing undesirable distributional consequences of taxation or financing other climate-related measures, including cuts in income and labor taxes and investment in public transportation. |

Sources: Author's assessments; IMF 2019; World Bank 2020, 2022.

In France, public backlash led to a suspension in the carbon tax increase in 2018. This experience highlights the political sensitivity of carbon pricing reforms and the need to address their poverty and inequality implications.

## Implications for Policy Design

While many countries have introduced or are planning to introduce carbon pricing schemes, the levels of carbon pricing have tended to be very low, with a number of countries taking a review-and-reevaluate approach. Climate actions are urgent, and countries need to take more ambitious steps in the fight against climate change. Therefore, climate policy design cannot wait for us to close knowledge gaps. The following general considerations may be helpful in designing strategies to mitigate the poverty and distributional impacts of carbon pricing, based on available evidence on the different impact channels and their attributes.

First, because the poverty and distributional impacts of carbon pricing are complex, reflecting and modeling all the impact channels in policy analysis is a challenge. The relative importance of each channel also likely depends on country-specific conditions and the designs of carbon pricing schemes. One viable strategy is to conduct a preliminary review of all the channels and pinpoint only the most salient ones in a detailed policy analysis and in the design of policy measures for any given country or region. However, the channels are often not independent of each other, and the modeling would need to take into account their linkages. For example, with partial pass-through, the portion that is passed backward in lower prices for firms would need to be distributed between capital and labor on the source side. Also, production responses by firms can alter factor prices for labor and capital in addition to reducing consumer prices.

Second, in the absence of mitigating measures, a carbon pricing reform may initially worsen poverty and inequality as the immediate effects dominate (e.g., through the consumption channel), before they gradually improve as the income channel starts to take effect. This suggests that the design of measures to mitigate poverty and distributional impacts should initially focus on short- to medium-term effects, which can help garner political support for carbon pricing reforms and protect the well-being of the most vulnerable. Such measures could be gradually phased out over time.

Third, policy designs should go beyond the averages. Large variations within income groups exist and may by associated with household characteristics, such as race and region. When administrative capacity allows, targeted measures should be considered to assist specific populations that are particularly affected.

Fourth, the findings may imply that the design of policy responses should start with the impact channels that are better understood and whose distributional effects are less uncertain, such as the consumption channel. Gradually, the design could be adjusted as new evidence emerges, including from reform experiences of other countries. The policy designs also may err on overcompensating rather than undercompensating the most vulnerable, given the uncertainties about the effects of some channels.

In addition to these guiding principles, the empirical literature on the poverty and distributional impacts of carbon pricing provides several main takeaways on how countries may move forward with their carbon pricing schemes:

1. The poverty and distributional impacts of carbon pricing vary by country and depend on the impact channels that are included in the analysis. Even when considering only the consumption channel with simplifying assumptions (e.g., full pass-through and no behavior responses by consumers and firms), the impacts vary significantly by country (Dorband et al. 2019; IMF 2019). In some cases, the income channel could fully offset the regressivity from the consumption channel (Rausch et al. 2010; Rausch, Metcalf, and Reilly 2011; Dissou and Siddiqui 2014; Beck et al. 2015; Goulder et al. 2019).

2. Many of the channels that are complex and difficult to model tend to lower the poverty and inequality impacts of carbon pricing, including behavior responses by consumers and firms, leakages, structural changes in factor income, and income and health gains from climate improvements. In addition, these channels often take time to materialize. Thus, the impact from not incorporating them initially is likely limited, particularly if a means-tested system is used to support the most vulnerable, as benefits are automatically adjusted with income, including from the carbon pricing channels.

3. Some groups, such as workers in carbon-intensive sectors, are particularly affected by carbon pricing. While the number of workers in these sectors may be relatively small, the impacts on the affected workers are large, and these workers tend to be vocal. Therefore, targeted measures, beyond government transfers (such as training and reemployment services and financial assistance related to job search and relocation), to assist these workers may be warranted, and the resource needs appear limited (Morris 2016; IMF 2019).

4. The revenue recycling channel is a key determinant of the poverty and distributional impacts of carbon pricing. Targeting the bottom 40 percent of the income distribution with around one-quarter of carbon revenues through government transfers appears sufficient to fully compensate them for rising energy prices (IMF 2019). However, political economy considerations may require the compensation of a larger share of the population, including through uniform transfers (Klenert et al. 2018). Policy designs also would need to balance efficiency objectives and equity objectives.

5. Overall, it is likely a reasonable starting point to base the analysis of the poverty and distributional impacts on the consumption channel with simplifying assumptions, targeted assistance to particularly affected workers, and targeted or universal cash transfers for households. In most cases, this design could fully protect low-income and vulnerable populations from carbon pricing reforms, while leaving a significant share of carbon revenues to improve economic efficiency.

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